

KHALED GALAL YEHIA

Software Engineer | Backend Development & AI Solutions

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience delivering enterprise-grade backend systems and AI-powered applications for banking and enterprise clients. Specialized in Java/Spring Boot, Python, and TypeScript, with hands-on delivery of Retrieval-Augmented Generation (RAG) pipelines, Large Language Model (LLM) integrations, REST API design, and workflow automation. Currently a member of the R&D AI Solutions team at Intercom Enterprises, building production tools adopted by QC, ECM, Finance, and Support teams company-wide. Pursuing an M.Sc. in Software Engineering.

TECHNICAL SKILLS

Programming Languages: Java, Python, TypeScript, JavaScript, SQL, C++

Backend Development: Spring Boot, Flask, Node.js, Struts 2, REST API Design, Microservices, Modular Monolith Architecture, System Integration

Frontend Development: Angular, TypeScript, JavaScript, HTML5, CSS3

AI & Machine Learning: Large Language Models (LLM), Retrieval-Augmented Generation (RAG), LangChain, Google Gemini, OpenAI GPT-4o, FAISS, pgvector, Vector Embeddings, Prompt Engineering, Machine Learning Models

Databases & Data Stores: PostgreSQL, Elasticsearch, Redis, pgvector, Vector Databases

DevOps & Tools: Docker, Git, Containerization, Monorepo Architecture

Enterprise Platforms: Flowable Workflow Engine, Appian BPM, UiPath Studio (RPA), IBM API Connect, IBM FileNet, OpenText ECM

PROFESSIONAL EXPERIENCE

Software Engineer, R&D AI Solutions

Intercom Enterprises | Cairo, Egypt | 01/2025 - Present

- Build AI-powered tools and enterprise applications that automate and streamline workflows for QC, ECM, Finance, and Support teams across the company.
- Automated the end-to-end QC workflow by developing QC Agent, a Flask and Angular monorepo deployed with Docker that generates test case titles from analysis documents using Google Gemini 2.0 Flash and OpenAI GPT-4o, applies embedding-based de-duplication, expands approved titles into detailed test steps, executes each scenario against the live system, and records test videos into a centralized run library.
- Designed and developed the Correspondence Management System (CMS), an enterprise modular monolith built with Spring Boot 4, Angular 21, PostgreSQL 16, and the Flowable workflow engine, digitizing company-wide correspondence processes.
- Delivered a production-ready Retrieval-Augmented Generation (RAG) agent using Flask, PostgreSQL with pgvector, and Google Generative AI, enabling employees to query internal policies and guidelines through a conversational interface.
- Automated technical documentation for the FileNet and OpenText ECM teams with a full-stack generator (Angular 18, Node.js, TypeScript, Google Gemini API) producing technical proposals, taxonomy definitions, and subscription event configurations, plus a Flask and LangChain FAISS-based RAG pipeline that drafts solution design documents from the same source analysis.
- Developed a bilingual (Arabic/English) desktop application in Python and Tkinter that automates the generation of financial proposals for the Finance team.
- Engineered a multi-layer log filtering system in TypeScript that isolates relevant entries from bank-submitted logs, accelerating issue investigation for first-line support engineers.

Backend Developer & Technical Support

Intercom Enterprises | Cairo, Egypt | 06/2024 - 12/2024

- Delivered backend development and production support across banking projects for United Bank, Banque du Caire, Arab International Bank, and Housing and Development Bank.
- Diagnosed and resolved production defects in mission-critical banking systems, maintaining the stability, performance, and availability of live customer-facing services.

Backend Developer

Intercom Enterprises | Cairo, Egypt | 10/2023 - 12/2023

- Developed backend services for the United Bank project while also contributing to front-end delivery, covering the full application stack.
- Improved transaction processing speed by implementing an instant payment process using Struts 2.

Digital Integration Trainee

Intercom Enterprises | Cairo, Egypt | 08/2023 - 09/2023

- Led development of a Blood System API on IBM API Connect, implementing accurate national ID validation rules.
- Built CRUD and MVC demo applications (online bookstore, student data management) and integrated OpenText for efficient document retrieval.

RPA Developer Trainee

Intercom Enterprises | Cairo, Egypt | 07/2022 - 09/2022

- Automated data extraction, form filling, and file transfer tasks using Robotic Process Automation (RPA) tooling.

BPM Developer Trainee

Intercom Enterprises | Cairo, Egypt | 02/2022 - 03/2022

- Designed and implemented a car mechanic workflow in Appian BPM that retrieves vehicle data and automatically routes repair information to mechanics.

EDUCATION

Master of Science in Software Engineering

Quantic School of Business and Technology | Washington, DC (Online) | 06/2025 - 08/2026 (Expected)

- DEAC-accredited program focused on AI-driven software engineering, system design, and modern development practices.

Bachelor of Science in Computer and Information Science

Ain Shams University | Cairo, Egypt | 09/2019 - 05/2023

- Graduation Project: Driver Behavior Detection System - designed and trained machine learning models to detect and classify unsafe driving behavior.

CERTIFICATIONS & TRAINING

- Digital Egypt Pioneers Program (DEPI) - Vulnerability Analyst / Penetration Tester, Ministry of Communications and Information Technology, Egypt (Infrastructure and Security), 11/2024 - 05/2025
- McKinsey.org Forward Program - problem solving, communication, adaptability, and digital toolkit, 12/2025
- Spring Boot 3 - backend development with the Spring Boot framework
- Comprehensive Cybersecurity Bootcamp
- RPA Developer Foundation (UiPath) | Appian Development (BPM)
- Front End Development Diploma | Cybersecurity Fundamentals Workshop